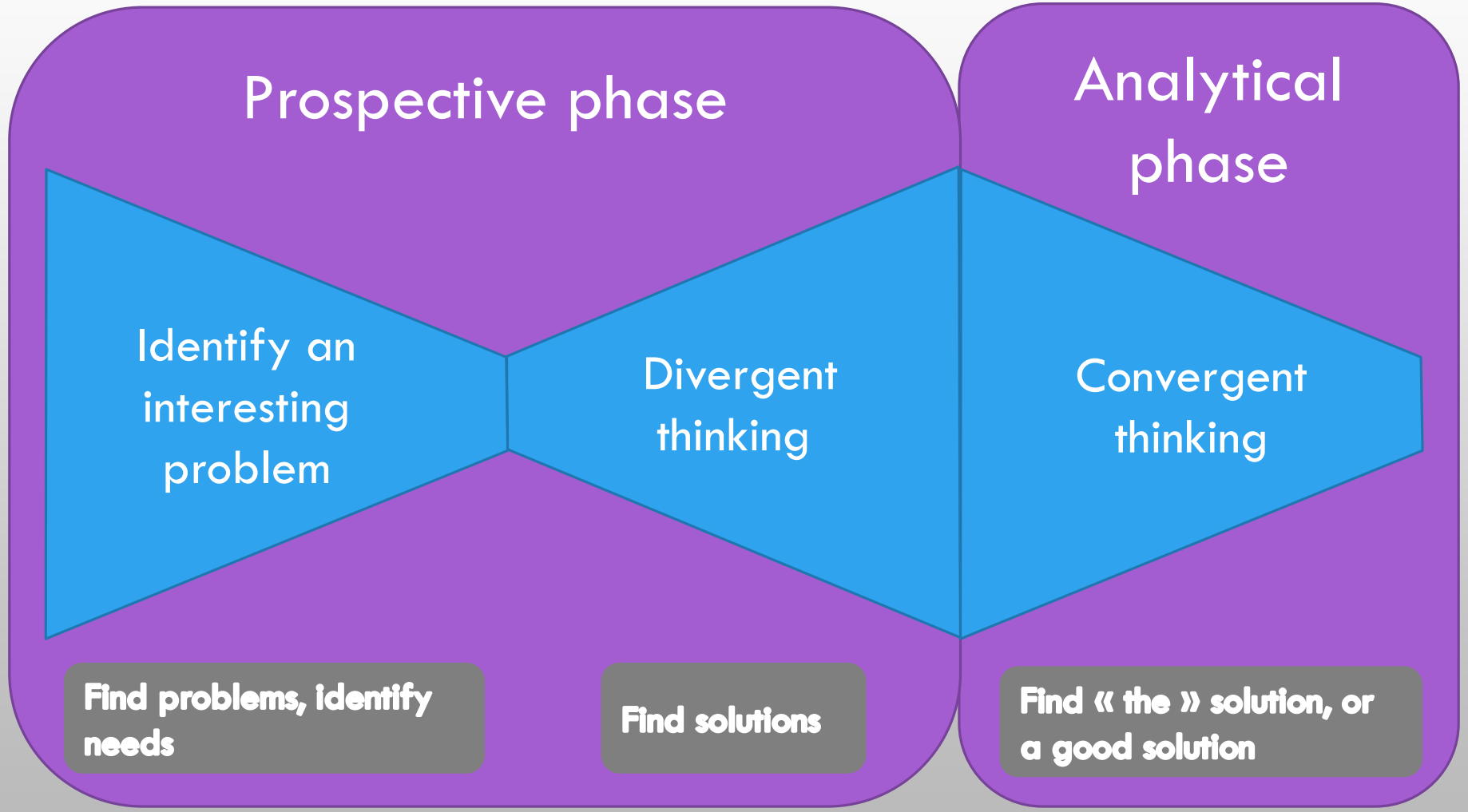
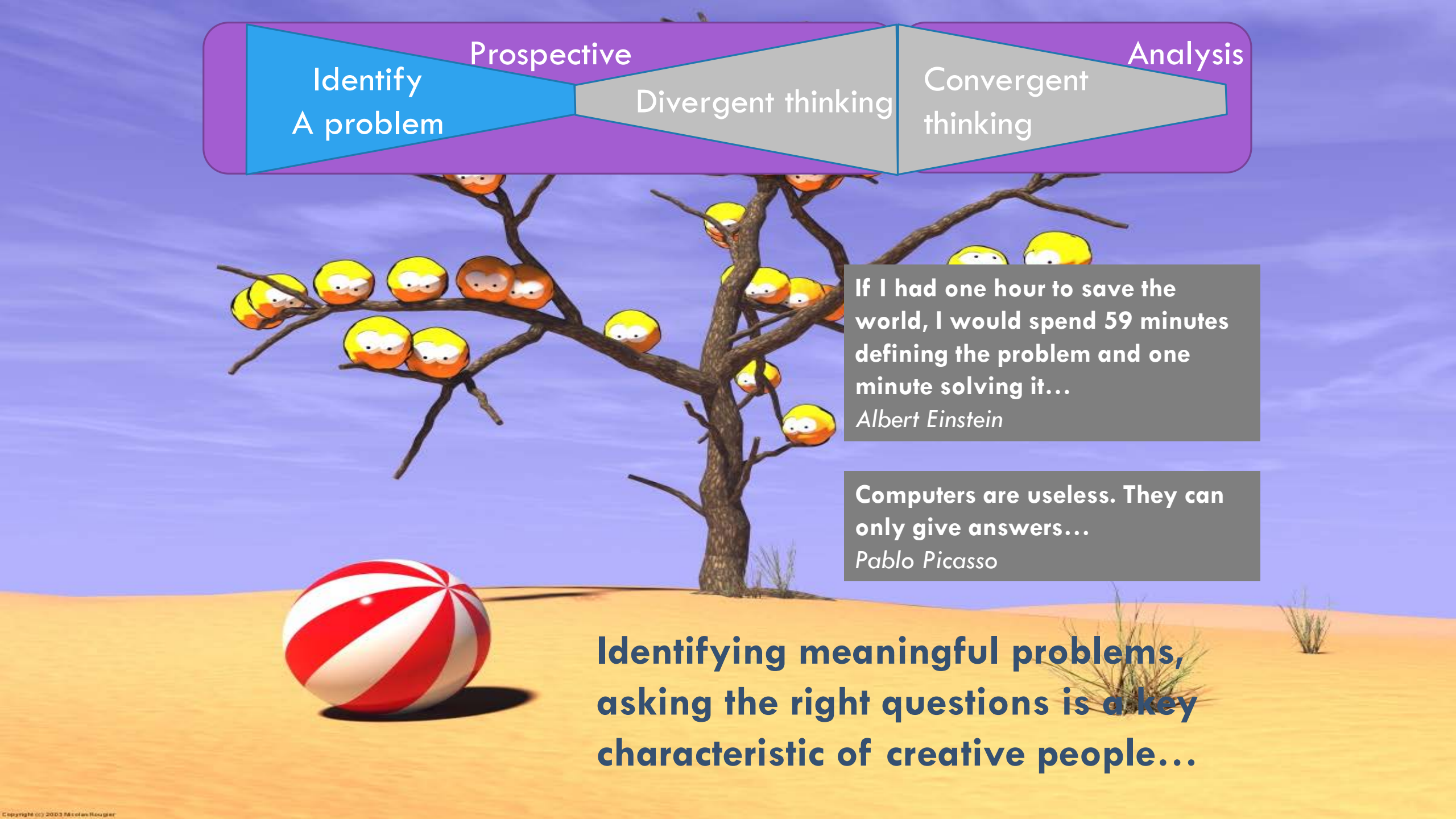
EXPLORING THE CREATIVITY PROBLEM

WEEK 2 MGMT2094

THE THREE STEPS OF CREATIVE PROBLEM SOLVING




Prospective
Identify
A problem
Divergent thinking
Analysis
Convergent
thinking
If I had one hour to save the world, I would spend 59 minutes defining the problem and one minute solving it…
Albert Einstein
Computers are useless. They can only give answers…
Pablo Picasso
Identifying meaningful problems, asking the right questions is a key characteristic of creative people…

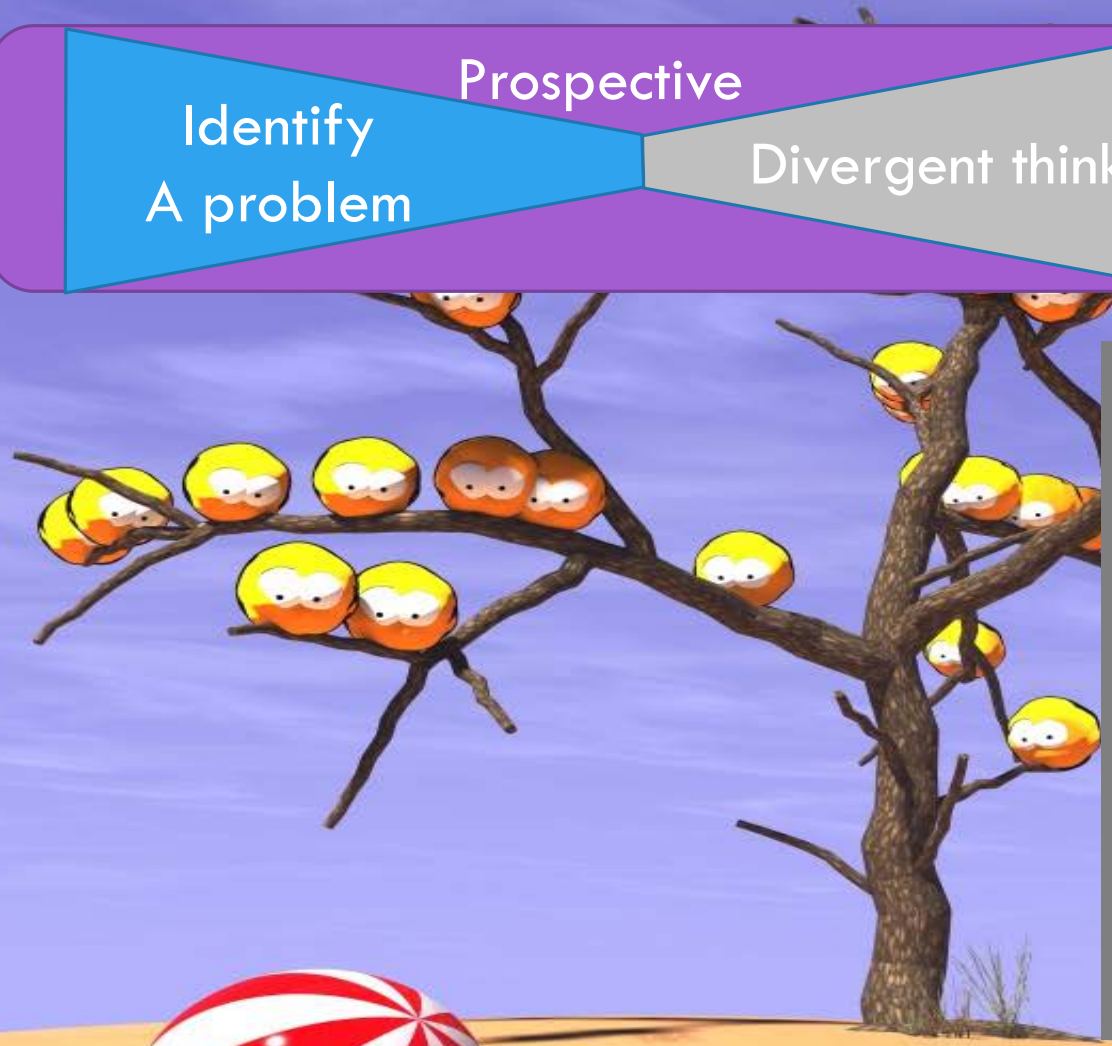
Prospective

Identify
A problem

Divergent thinking

Analysis

Convergent
thinking



Examples of important questions :

Why is the speed of light the same everywhere ?... led to the discovery of relativity by Einstein

How to use colours differently?... one of the essential questions that led to impressionism

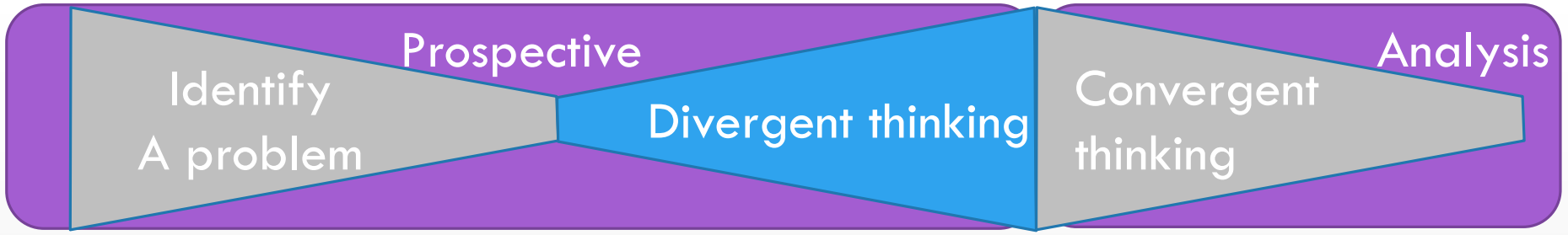
How to make bus stop shelters viable?... led to the worldwide success of Decaux



Some related qualities :

- **Listen to the unusual**
- **Identify consequences that make the subject essential**

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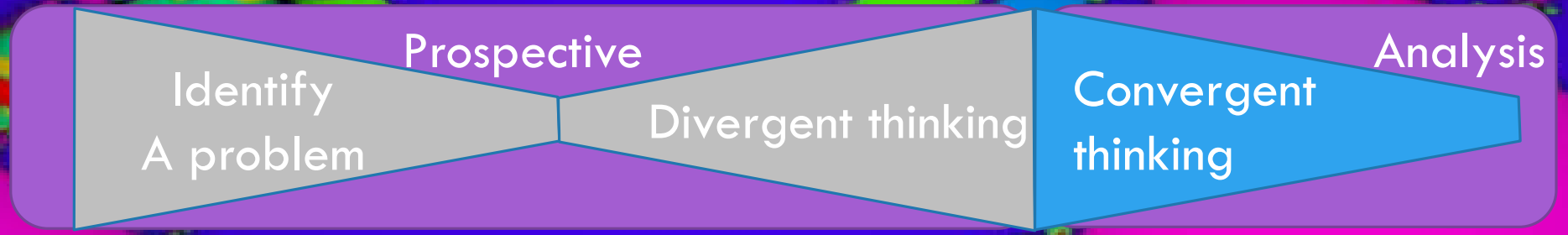


Example : the quota method

Always ask at least three ideas from your team (& yourself!):
- Different options, alternate solutions
>> Or refuse to decide

When at least three good ideas have been found, many more are generally found!

Break the "linear thinking" propensity to :
- settle for the first idea
- logical reasoning, that locks you in the first solution



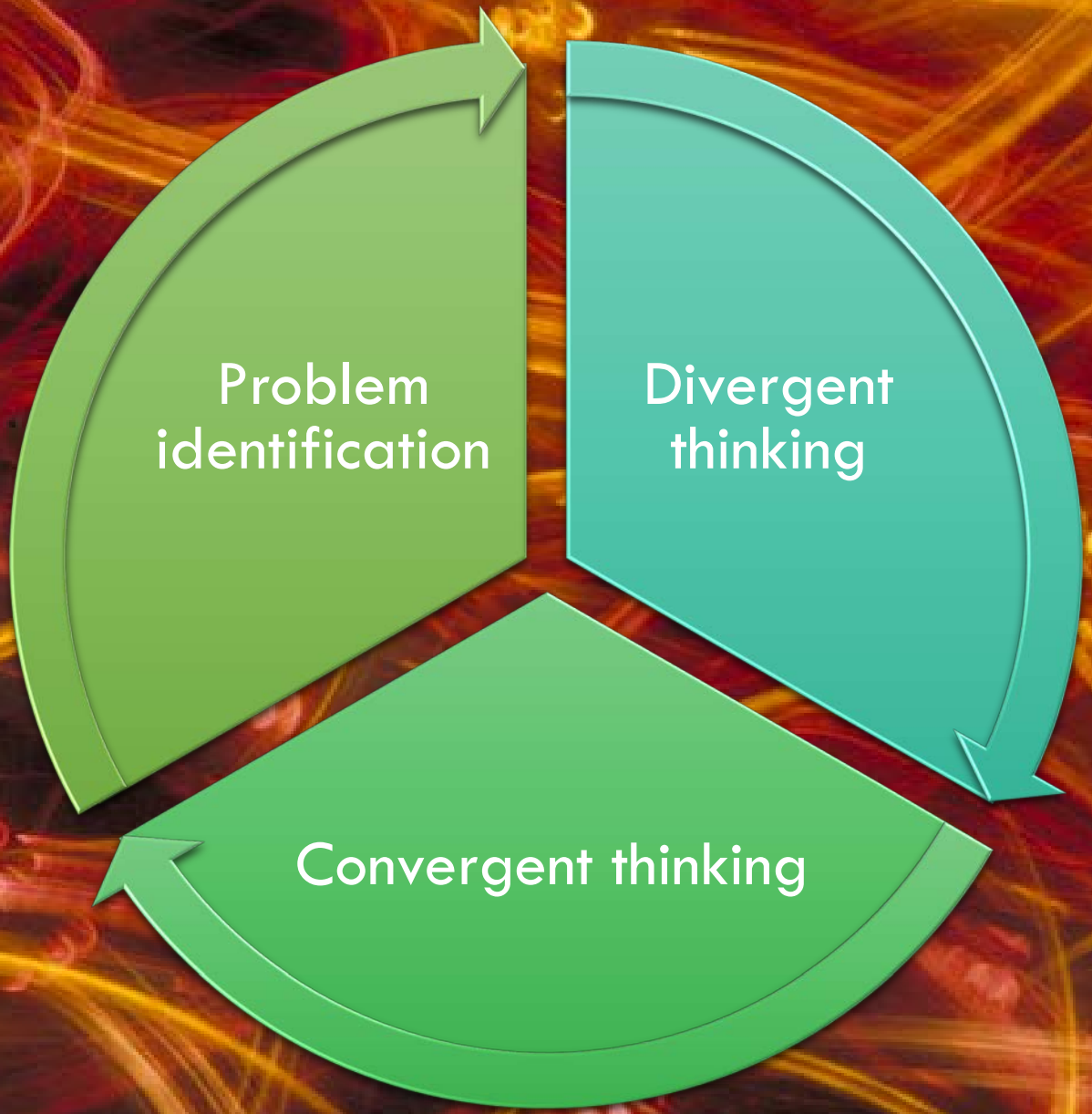
No standard test

Ability to identify benefits and risks. Keep the best and eliminate wrong answers. Evaluation / selection.

Positive and critical thinking

IQ Tests

Deductive capacity: deduce the consequences.



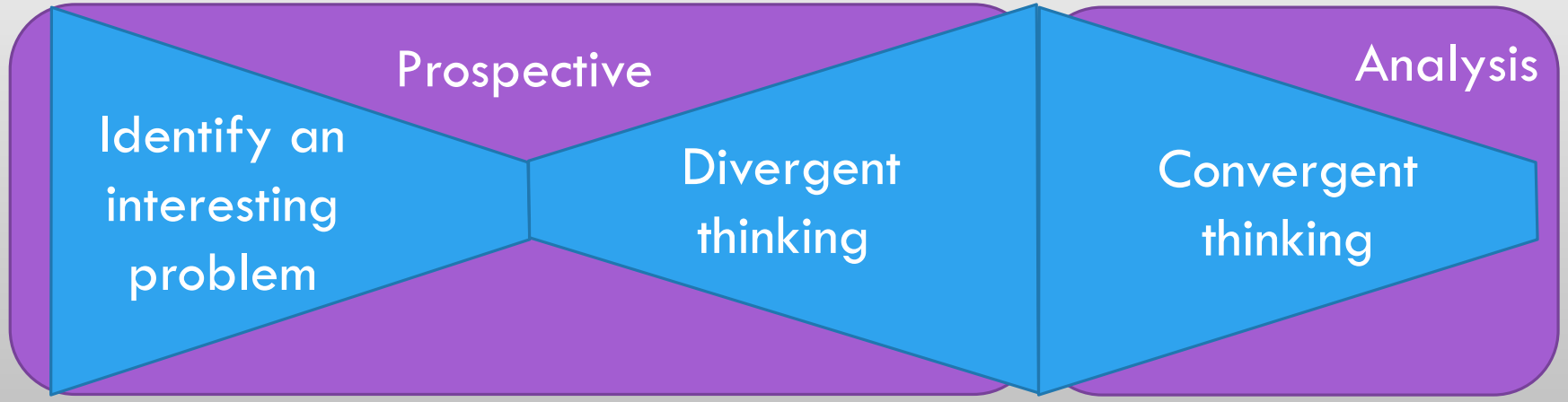
Great advances are usually the result of a long process that is a sum of intuitions and successive progresses.

WHY CANT WE CREATE A STANDARDIZED TEST MEASURING INDIVIDUAL CREATIVITY?

Creativity is a process

Creativity is often a group phenomenon

Creativity implies domain knowledge



No standard test for ability to formulate interesting problems

Divergent thinking tests produce mixed results

IQ tests

Emotional Intelligence

CREATIVITY TECHNIQUES

Rational « Serious creativity »

- TRIZ *Genrich Altshuller*
- Lateral thinking, 6 hats *Edward De Bono*
- Mindmaps *Tony Buzan*

Gaming

- Frame games *Thagi*
- Improvisation
- Innovation games *Luke Hohmann*
- Constructions: LEGOs, blocs…

Idea gathering

- Brainstorming, brainwriting, method 635, CPS
- Idea box, idea intranets, participatory innovation
- Crowdsourcing

Irrational

- Detour approaches
- Synectics (analogical reasoning)
- Random approaches

IDEA GATHERING TECHNIQUES

Brainstorming, brainwriting methods



Alex Orborn

President & co-founder of BBDO

In order to answer a question, generate in a meeting as many ideas as possible in a short time without exercising censorship on the participants.

Analyze and prioritize ideas afterwards.

Participatory innovation Idea boxes, idea intranets



1721: the best ideas deposited in a “idea box” (meyasu bako) are rewarded.

Shôgun Tokugawa Yoshimune

Image : en.wikipedia.org/wiki/File:Tokugawa_Yoshimune.jpg

Seventies : japanese companies develop quality circles and « Kaizen ».

2000s: idea management software, participatory innovation: open procedures for the collection, analysis and application of ideas.

2010s: companies seeking new ideas outside with idea crowdsourcing : Crossing, YourEncore, CrowdSpirit, InnoCentive

RATIONAL APPROACHES



Edward de Bono

Lateral thinking, 6 hats…

"Lateral thinking" escapes from the linear thinking for example by introducing a random component in reasoning.

The "six hats" are a means to materialize the most common reasoning attitudes.

Mindmaps



Tony Buzan

Image : www.tonybuzan.com/gallery/photos/

Means to visually represent ideas, facts …

Mindmaps are used to summarize existing knowledge, gather ideas (brainstorming), or organize tasks or information (meeting management…).



TRIZ



Genrich Altshuller

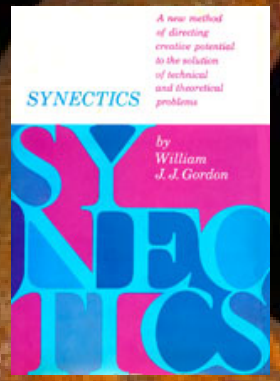
Image: commons.wikimedia.org/wiki/File:Genrikh_Saulovich_Altshuller_commemorative_plaque_in_Petrozavodsk.JPG

He discovers universal laws of technological invention. Each solves a contradiction between physical quantities.

To innovate, one identifies contradictions, then applies these laws, which provide avenues to explore.



« NON RATIONAL » TECHNIQUES



William Gordon

Analog techniques shift the problem with analogies, and explores them.

"Making the familiar strange, making the strange familiar." 3 types of analogies:

- Direct: natural (animal, plant), or graphic
- Symbolic parable, metaphor
- Fantasy: dream, daydream

Personal approach: personal identification with the situation

Detour techniques : one distances themself from the problem and then they return to it to converge. The distance stimulates imagination. Idea production follows.

Distanciation phase

Principle : distort and transform the problem into an evocation that stimulates the imagination

Approaches :

Deformation, forced encounters, dreamlike techniques, projective graphics …

Ideation phase

1 / Blanks fuzzy, ambiguous ideas,

2 / Scanning / crossover: one follows the track to push the idea as far as possible until it becomes useful.

Principle : turn ideas into positive. To make it work, it would have to …

4 METHODS USEFUL FOR MARKETING & SALES

Mindmaps



Tony Buzan

Represent information (ideas or facts) in the form of a diagram called "mental map"

Usage: synthesis of information / collaborative brainstorming ...

Innovation games



Luke Hohmann

Fun activities rather than games rather, they aim to collect, organize and prioritize ideas

Usage : product roadmap, identifying growth drivers ...

Lateral thinking, 6 hats

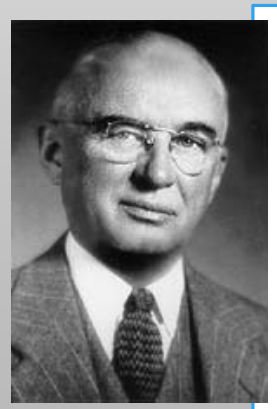


Edward de Bono

Approach to overcome the limitations of the "linear, logic thinking" while remaining rational

Usage : problem solving, product ideas ...

Brainstorming



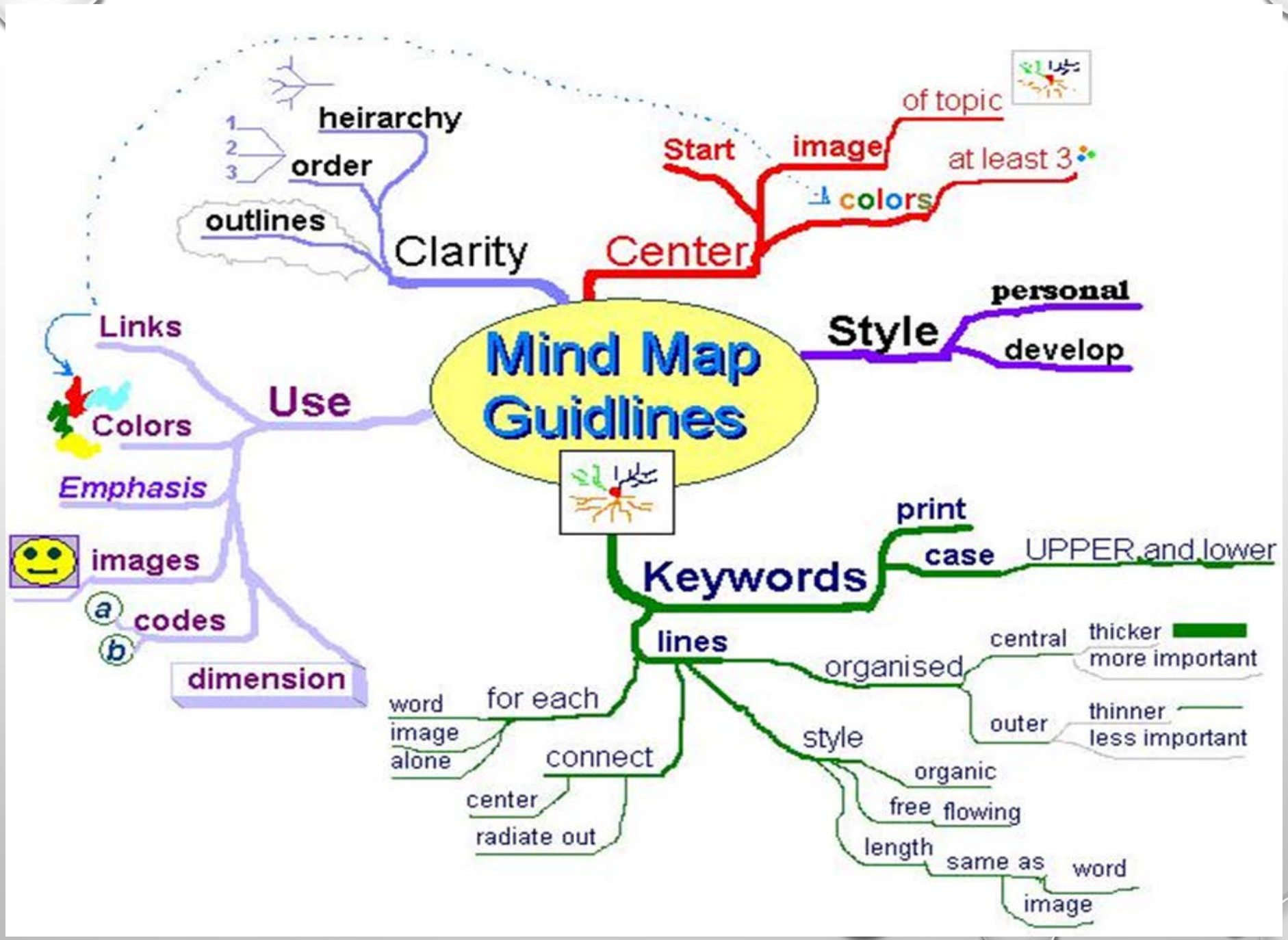
Alex Osborne (BBDO)

Collection of ideas to produce a maximum of ideas which are then analyzed and sorted

Usage : product ideas, solutions to a problem, names, slogans ...

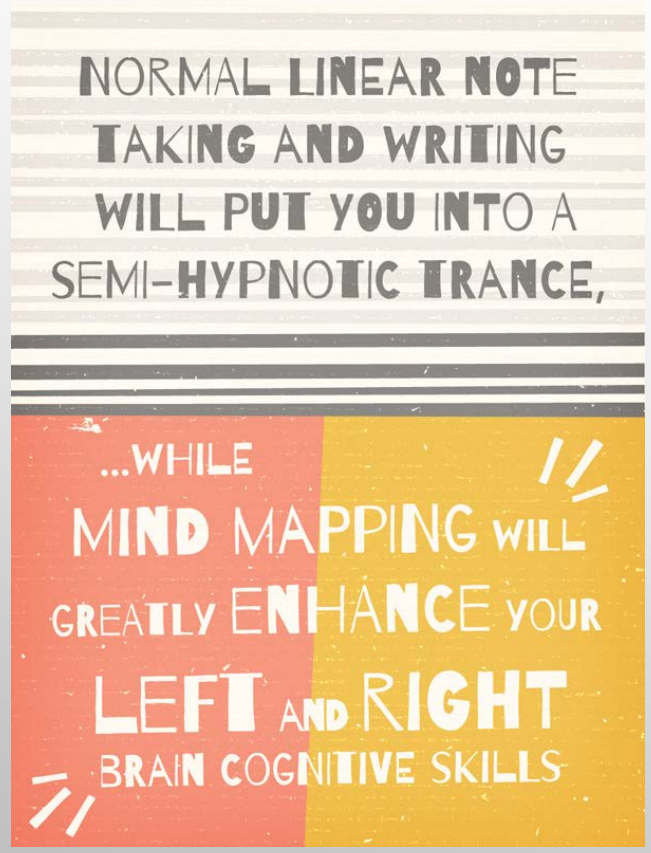
WHAT ARE MIND MAPS?

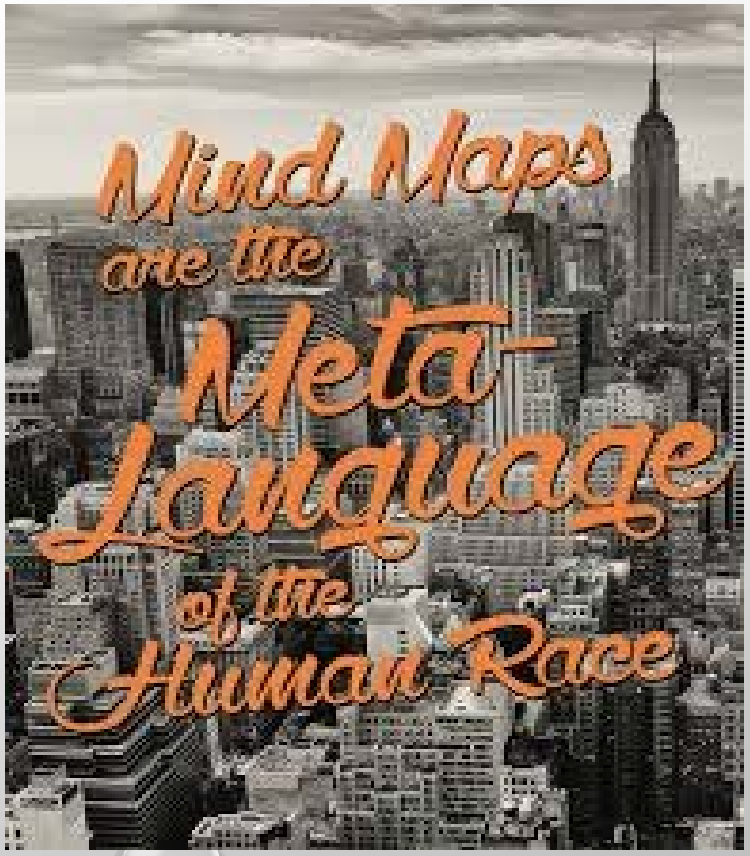
- a mind map is a way to visually organize notes, thoughts, ideas, and conversations.
- Tony Buzan takes the credit for inventing modern mind mapping. (a british psychologist)
- in reality different forms of mind mapping have been used over many centuries – buzan really just popularized it and offered some theories around why they work so well to capture thoughts and are compatible with the thought process.
- a mind map is created around a single word or text, placed in the centre, to which associated ideas, words and concepts are added.
- mind maps have many uses including business situations where you could use mind maps to document conversations, general note taking, brainstorming, or to tweeze out a strategy or a particularly complicated situation you may face.
- in business mind maps are most frequently used collaboratively – but it is possible to use mind mapping on your own, as an individual.
- See handout on Mindmap guidelines on ilearn. Tony Buzan brief overview of mindmapping below.
- https://www.youtube.com/watch?v=MlabrWv25qQ


Mind Map Guidlines
Clarity
heirarchy
order
outlines
Center
Start
image
of topic
at least 3
colors
Style
personal
develop
Use
Links
Colors
Emphasis
images
codes
dimension
Keywords
print
case
UPPER and lower
lines
organised
central
thicker
more important
outer
thinner
less important
for each
word
image
alone
connect
center
radiate out
style
organic
free flowing
length
same as
word
image

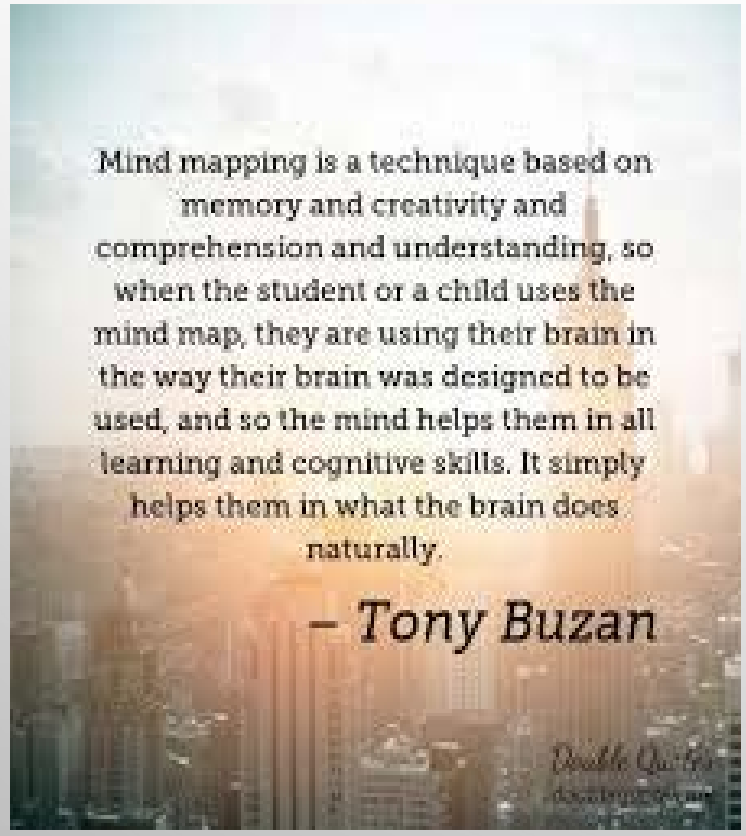

FINANCIAL PLANNING
FOR SMALL BUSINESS OWNERS
Business owner
V. self-employed
E-Myth book
Setting up systems
business planning
marketing plans
SWOT analysis
Selling your business
who are likely buyers?
how to get the best price?
buy-sell agreements
Saving for your retirement
how much do you need?
your business
borrowing money for your business
banks
mortgage brokers
other lenders
insurance
personal
business
Setting up your business
company
directors' liabilities
partnership
sole trader
Investments
Kiwisaver
Do-it-yourself investing
managed funds
using a stock broker
fixed interest investments

QUOTES


NORMAL LINEAR NOTE TAKING AND WRITING WILL PUT YOU INTO A SEMI-HYPNOTIC TRANCE,
...WHILE MIND MAPPING WILL GREATLY ENHANCE YOUR LEFT AND RIGHT BRAIN COGNITIVE SKILLS


Mind Maps are the Meta Language of the Human Race

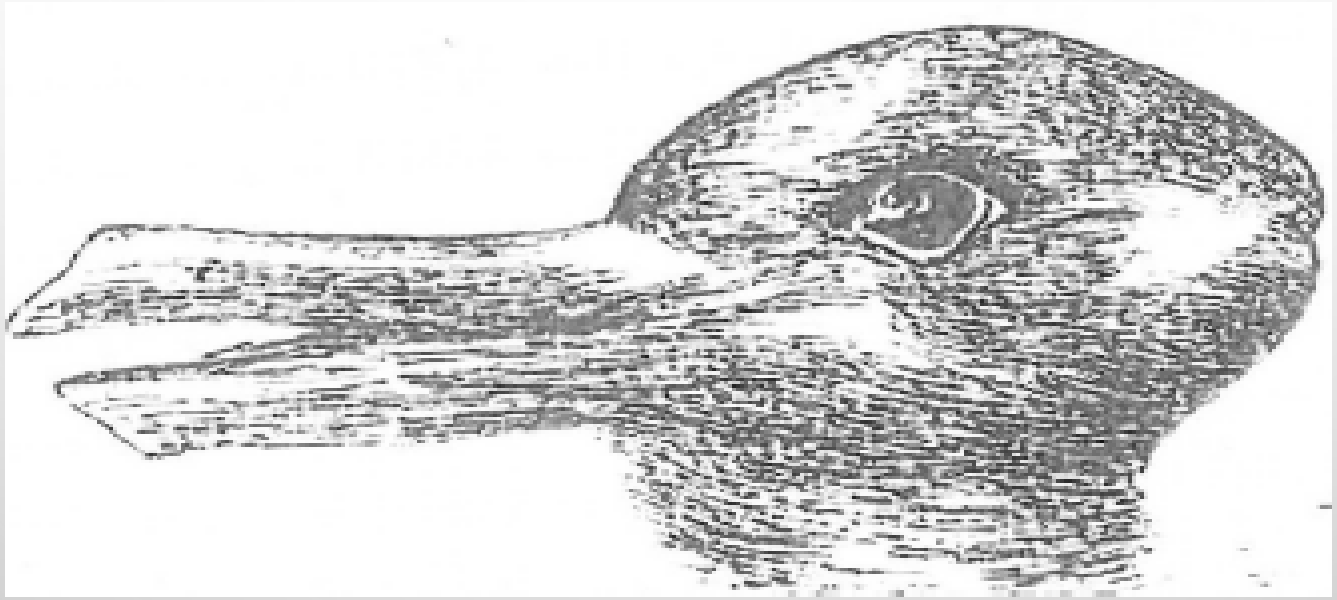
MORE QUOTES ON MIND MAPPING






PLEASE
DO
NOT
DISTURB
MIND MAPPING
THANK YOU

CREATIVITY TEST



TESTS OF CREATIVITY.

- HOW EASY WAS IT TO SEE BOTH?
- DID IT REQUIRE REAL MENTAL STRAIN?
- THE EASE TO FLIP BETWEEN THE TWO IS RELATED TO CREATIVITY. (WISEMAN ET AL. 2011)

- NOW LETS THINK ABOUT A DOOR?
- HOW MANY USES CAN YOU WRITE DOWN IN ONE MINUTE?

NEXT WEEK